

ETL Pipeline & Data Integration Design

Milestone 1 – Task 5: Data Integration Pipeline Design

Prepared By:

Lakshmikanth Reddy

Organization:

Infosys Springboard Internship Program

Project:

AI-Powered Workforce Analytics & Talent Intelligence Dashboard

Table of Contents

- 1. Introduction
- 2. Enterprise ETL Pipeline Design
- 3. Data Flow Diagram (DFD)
- 4. Data Integration Architecture
- 5. Integration Planning Workflow
- 6. Conclusion

1. Introduction

In today's world, a large amount of data is generated every day. Before this data can be used for analysis or decision-making, it needs to be organized and cleaned. This process helps remove errors, fill missing information, and make the data easier to understand.

In this project, an HR Employee dataset is used to learn the basic steps of handling data. The work includes collecting the data, cleaning it, making necessary changes, and saving it in a proper format. These steps are known as the ETL (Extract, Transform, and Load) process.

The main goal of this milestone is to understand how raw data is converted into clean and organized data that can be used for reports, analysis, and future projects. This milestone also includes simple diagrams to explain the flow of data from the source to the final output.

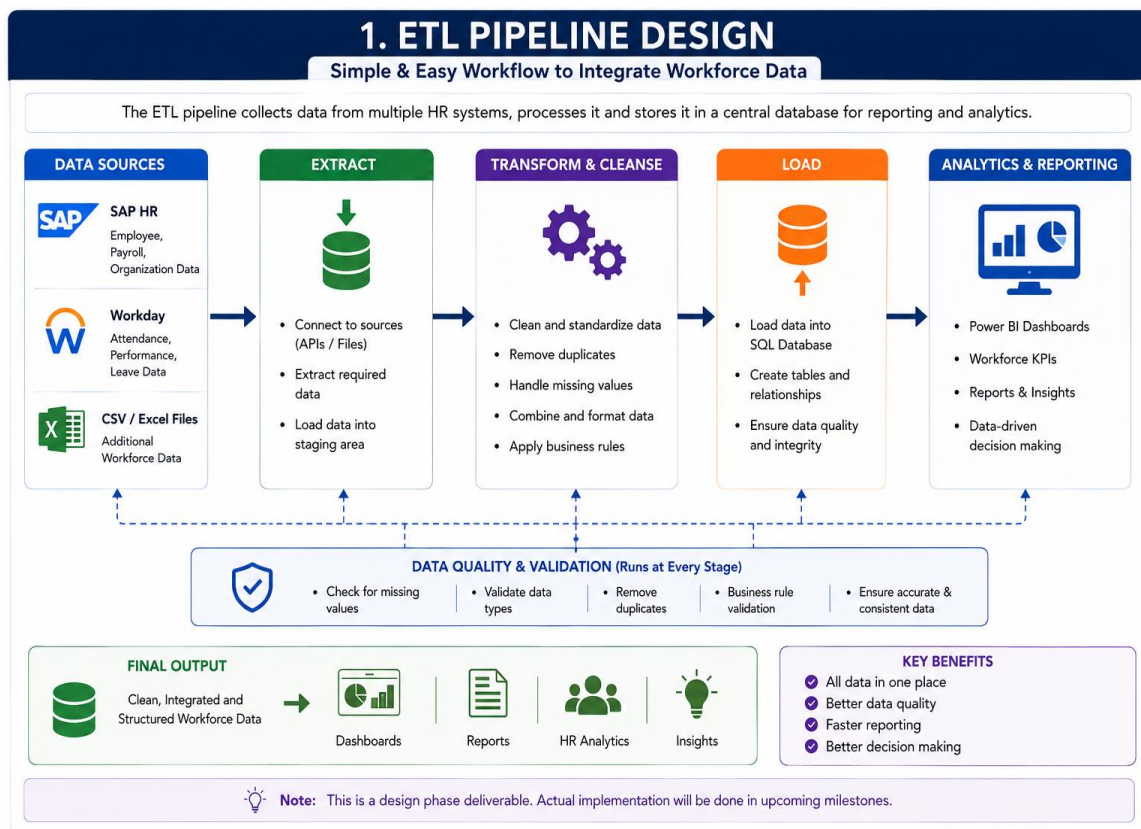
2. Enterprise ETL Pipeline Design

Purpose:

Design the complete Extract, Transform and Load workflow.

Explanation:

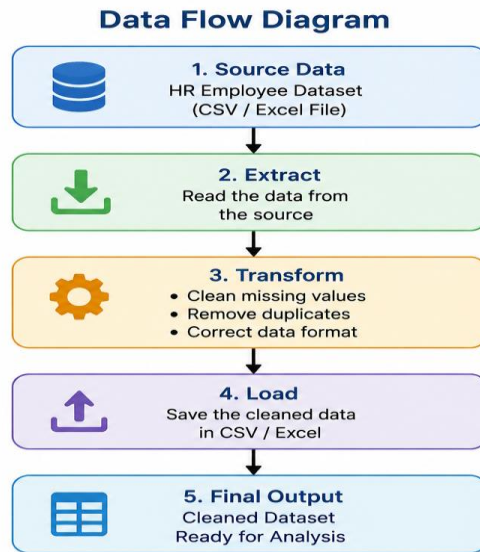
- Extract workforce data from SAP HR, Workday and CSV/Excel files.
- Validate data quality.
- Transform and standardize datasets.
- Load into SQL database.
- Generate dashboards and analytics.



3. Data Flow Diagram (DFD)

Purpose:

Illustrates how data moves from data sources to end users through the ETL process.



The Data Flow Diagram shows how the HR Employee data is processed step by step.

1. Source Data:

The process starts with the HR Employee dataset, which is stored in a CSV or Excel file. This file contains the raw employee information.

2. Extract:

In this step, the dataset is read from the source file so that it can be processed.

3. Transform:

The extracted data is cleaned and organized. Missing values are handled, duplicate records are removed, and incorrect data formats are corrected. This makes the data accurate and ready for use.

4. Load:

After cleaning, the processed data is saved into a new CSV or Excel file. This becomes the final cleaned dataset.

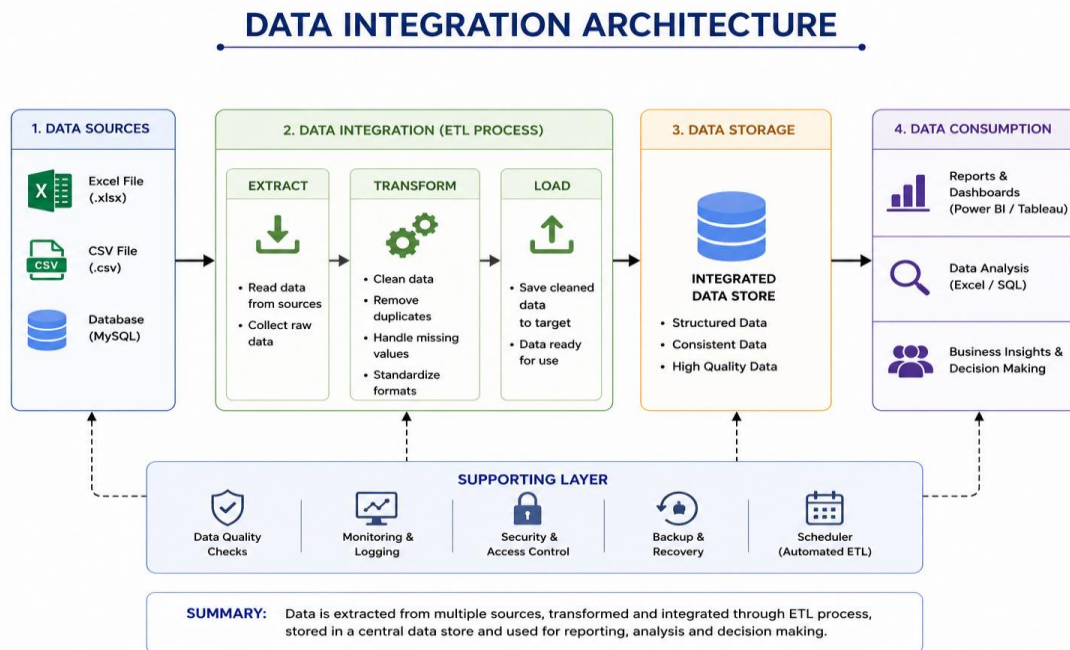
5. Final Output:

The cleaned dataset is now ready for analysis, reporting, or visualization. Since the data is organized and error-free, it can be used to generate meaningful insights.

4. Data Integration Architecture

Purpose:

Illustrates the layered architecture of the workforce analytics platform.



The Data Integration Architecture shows how data is collected, processed, stored, and used for analysis.

1. Data Sources:

The process begins with different data sources such as Excel files, CSV files, and databases. These sources contain the original or raw data.

2. Data Integration (ETL Process):

The ETL process has three main steps:

- **Extract:** Data is collected from different sources.
- **Transform:** The collected data is cleaned by removing duplicate records, handling missing values, and correcting data formats.
- **Load:** The cleaned data is stored in a central location for future use.

3. Data Storage:

The processed data is saved in an integrated data store. This ensures that the data is organized, accurate, and ready for analysis.

4. Data Consumption:

The stored data is used to create reports, dashboards, and perform data analysis. These insights help organizations make better business decisions.]

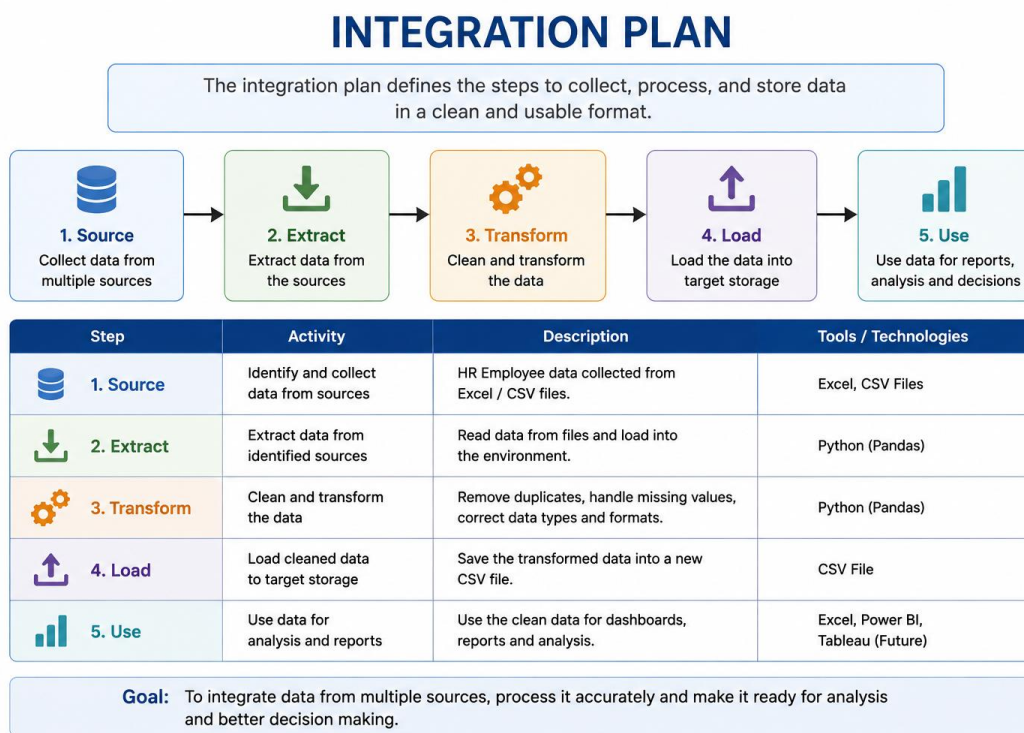
5. Supporting Layer:

The supporting layer includes data quality checks, monitoring, security, backup, and scheduling. These features help keep the data safe, reliable, and available whenever it is needed.

5. Integration Planning Workflow

Purpose:

Shows the sequence of planning activities required before implementation.



The Integration Plan describes the steps followed to collect, process, and use the data. It helps ensure that the data is accurate, organized, and ready for analysis.

1. Source:

The first step is to collect data from different sources such as Excel files, CSV files, or databases.

2. **Extract:**

The required data is extracted from these sources and prepared for processing.

3. **Transform:**

The extracted data is cleaned by removing duplicate records, filling or handling missing values, and correcting data formats. This improves the quality of the data.

4. **Load:**

The cleaned data is saved into a new file or storage location so that it can be used easily.

5. **Use:**

Finally, the processed data is used to create reports, dashboards, and perform data analysis. This helps users understand the data and make better decisions.

Goal:

The main goal of the integration plan is to convert raw data into clean and organized data that is ready for analysis and reporting.

6.Conclusion

In this milestone, the basic concepts of data integration and the ETL process were successfully understood and implemented. The HR Employee dataset was collected, cleaned, transformed, and stored in a structured format. Simple diagrams such as the Data Flow Diagram, Data Integration Architecture, and Integration Plan were created to explain how the data moves through each stage of the process.

This project helped in understanding how raw data is converted into clean and organized data that is ready for analysis. The knowledge gained from this milestone provides a strong foundation for future work in data analysis, reporting, and data engineering.